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Experiment Number : 09

Experiment Title : Generation and Observation of Amplitude Modulated (AM) Signals

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Experiment No. 09

Title: Generation and Observation of Amplitude Modulated (AM) Signals

Aim

To design and simulate an Amplitude Modulation (AM) circuit in LTspice using a Behavioral Voltage Source acting as a multiplier. To observe the modulated waveform in the time domain and calculate the modulation index for three different modulating signal amplitudes.

Software Used

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory

1. What is Modulation?

Modulation is the process of varying a property of a **high-frequency carrier signal** according to the **low-frequency message signal (information signal)** in order to transmit the information efficiently over long distances.

The carrier signal can have its:

- **Amplitude**
- **Frequency**
- **Phase**

changed depending on the type of modulation.

Why modulation is needed

- Enables long distance transmission 📡
- Reduces antenna size
- Avoids mixing of signals
- Improves signal quality

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2. What is Amplitude Modulation (AM)?

Amplitude Modulation is a type of modulation in which the **amplitude of the carrier wave is varied** in proportion to the **instantaneous amplitude of the message signal**, while **frequency and phase remain constant**.

$$y = A_c\left(1 + m\cos\left(\omega_m t\right)\right)\cos\left(\omega_c t\right)$$

Where:

- A_c= amplitude of carrier wave
- m= modulation index
- ω_m= angular frequency of message signal
- ω_c= angular frequency of carrier signal

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3. What is Modulation Index?

Modulation index (m) indicates the **extent to which the carrier amplitude is varied** by the modulating signal.
It is defined as the ratio of amplitude of modulating signal to amplitude of carrier signal.

$$m = \frac{A_m}{A_c}$$

Where:

- A_m= amplitude of message signal
- A_c= amplitude of carrier signal

It can also be calculated from the waveform:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

Value of modulation index:

- $0 \, lemle \, 1 \rightarrow$ proper modulation
- $m > 1 \rightarrow$ distortion occurs

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4. Under Modulation, Critical Modulation and Over Modulation
a) Under Modulation
When the modulation index is **less than 1**

$$m < 1$$

- Carrier amplitude is not fully utilized
- Signal strength is weaker
- No distortion occurs

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b) Critical Modulation (100% Modulation)
When the modulation index is **equal to 1**

$$m = 1$$

- Maximum possible modulation without distortion
- Best signal strength
- Ideal condition

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c) Over Modulation
When the modulation index is **greater than 1**

$$m > 1$$

- Envelope gets distorted
- Information cannot be recovered properly
- Causes signal interference

Mathematical Representation of AM

Let:

- Carrier signal: $c(t) = A_c \sin(2\pi f_c t)$
- Message signal: $m(t) = A_m \sin(2\pi f_m t)$

The standard expression for an AM signal is represented as:

$$s(t) = [A_c + m(t)] \sin(2\pi f_c t) \qquad \text{.....} \quad (1)$$

where

- A_m is the peak amplitude of the modulating signal
- A_c the peak amplitude of the carrier signal

The modulation index is defined as the ratio of the peak amplitude of the modulating signal (A_m) to the peak amplitude of the carrier signal (A_c).

$$\text{Modulation index } (\mu) = \frac{A_m}{A_c}$$

For simplicity, we can take $A_c = 1$, so that A_m will directly represent modulation index. Thus, the AM signal can be given by:

$$s(t) = [1 + m(t)] \sin(2\pi f_c t) \qquad \ldots. \ (2)$$

Circuit Implementation of the AM Equation

The AM equation $s(t) = [A_c + m(t)] \sin(2\pi f_c t)$ has three distinct parts:

Part 1 – [Ac + m(t)]: A DC value Ac (=1) added to the message signal m(t). This is realised by connecting one DC source (Ac = 1) and one SINE voltage source (message m(t)) in series (A_m value will decide the modulation index).

Part 2 – sin ($2\pi f_c t$): Multiplication by the normalised carrier c(t) with $A_c = 1$. This is realised by a separate SINE voltage source for the carrier.

Part 3 – multiplication of Part 1 and Part 2: This can be realised by a Behavioral Voltage Source multiplier.

Component	Source Type	Signal Produced	Key Setting
V1	SINE source	Normalised carrier: $\sin(2\pi \cdot f_c \cdot t)$	Amplitude = 1V, Freq = 100k, DC offset = 0
V2	SINE source	Message signal: $A_m \cdot \sin(2\pi \cdot f_m \cdot t)$	Amplitude = Am (varies per case from 0 to 2), Freq = 1k, DC offset = 0
V3	DC source	Bias: Ac = 1V	DC Value = 1 (equals A_c)
B1	Behavioral Voltage (BV)	AM output: V(sum) × V(carrier)	V = V(sum) * V(carrier)
R1	Resistor	50Ω output load	Value = 50

Simulation Procedure

Step 1: Place the Normalised Carrier Source Vc (SINE, 100 kHz, Vamp = 1)

- Select voltage source, and place it.
- Right-click on the voltage source → Advanced → select SINE.
- Enter the following values:

```
DC offset (Voffset) = 0
Amplitude (Vamp) = 1          ← Unit amplitude — normalised carrier
Frequency = 100k              ← fc = 100kHz (carrier frequency)
Tdelay = 0, Theta = 0, Phi = 0
```

- Right-click on the label V1 and replace it by Vc
- Press 'G' → place ground below Vc.
- Press 'W' to wire the components. Connect negative (–) terminal of Vc to Ground.
- Connect a free wire from positive (+) terminal of Vc
- Press 'N' for assigning Net Name. Type 'carrier' and press Ok.
- Paste it on the free wire.
- Hover over the free wire — bottom status bar must show 'carrier' (not N001 or N002).

Step 2: Place the Message Signal Source Vm (SINE, 1 kHz)

- Select voltage source, and place it to the right to voltage source Vc.
- Right-click on the voltage source → Advanced → select SINE.
- Enter the following values:

```
DC offset (Voffset) = 0
Amplitude (Vamp) = 0.5        ← Am = 0.5 (change for other cases)
Frequency = 1k                ← fm = 1kHz (message frequency)
Tdelay = 0, Theta = 0, Phi = 0
```

- Right-click on the label V2 and replace it by Vm

Step 3: Place the DC Bias Source V3 (Ac = 1V DC)

- Select voltage source, and place the third source below Vm
- Right-click V3 → in the simple Value field type:

  1 ← 1V DC = Ac (carrier amplitude)

- Press 'G' → place ground below V3.
- Press 'W' to wire the components. V3 negative (–) → GND
- V3 positive (+) → Vm negative (–) ← wire these two pins together
- Vm positive (+) → free wire
- Press 'N' for assigning Net Name. Type 'sum' and press Ok.

- Paste it on the free wire.
- Hover over the free wire — bottom status bar must show 'sum' (not N001 or N002).

Step 4: Place the Behavioral Voltage Source B1 (Multiplier)

- Press 'P' → search bv → select Behavioral Voltage → place to the right of Vc, Vm, V3.
- Right-click B1 → in the Value field type EXACTLY:

  V = V(sum) * V(carrier)

B1 Expression — Verification:

```
V(sum) = Ac + Am·sin(2π·fm·t)          ← from V3 + Vm series (Step 3)
V(carrier) = sin(2π·fc·t)              ← from Vc with Vamp=1 (Step 1)
B1 output = V(sum) × V(carrier)
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= [Ac + Am·sin(2π·fm·t)] × sin(2π·fc·t)

= s(t) matches standard AM equation (1)

Step 5: Place Load Resistor R1 and Complete the Circuit

- Press 'R' → place resistor. Right-click → set value to 50.
- Press 'W' → connect B1 positive (+) terminal to top of R1.
- Press 'N' → label this wire: AM_out
- Press 'G' → ground B1 negative (−) terminal and R1 bottom terminal.

Step 6: Set Up and Run Transient Simulation

- Click Simulate → Edit Simulation Cmd → Transient tab.
- Set Stop Time:
  - .tran 5m ← 5 milliseconds = 5 complete message cycles at 1 kHz
- Place this directive on the schematic canvas. Click Run.

Step 7: View Waveforms

- Click on the AM_out wire → adds V(AM_out) to the plot.
- Click on the sum wire → adds V(sum) to see [Ac + m(t)] envelope.
- Click on the carrier wire → adds V(carrier) to see normalised carrier.

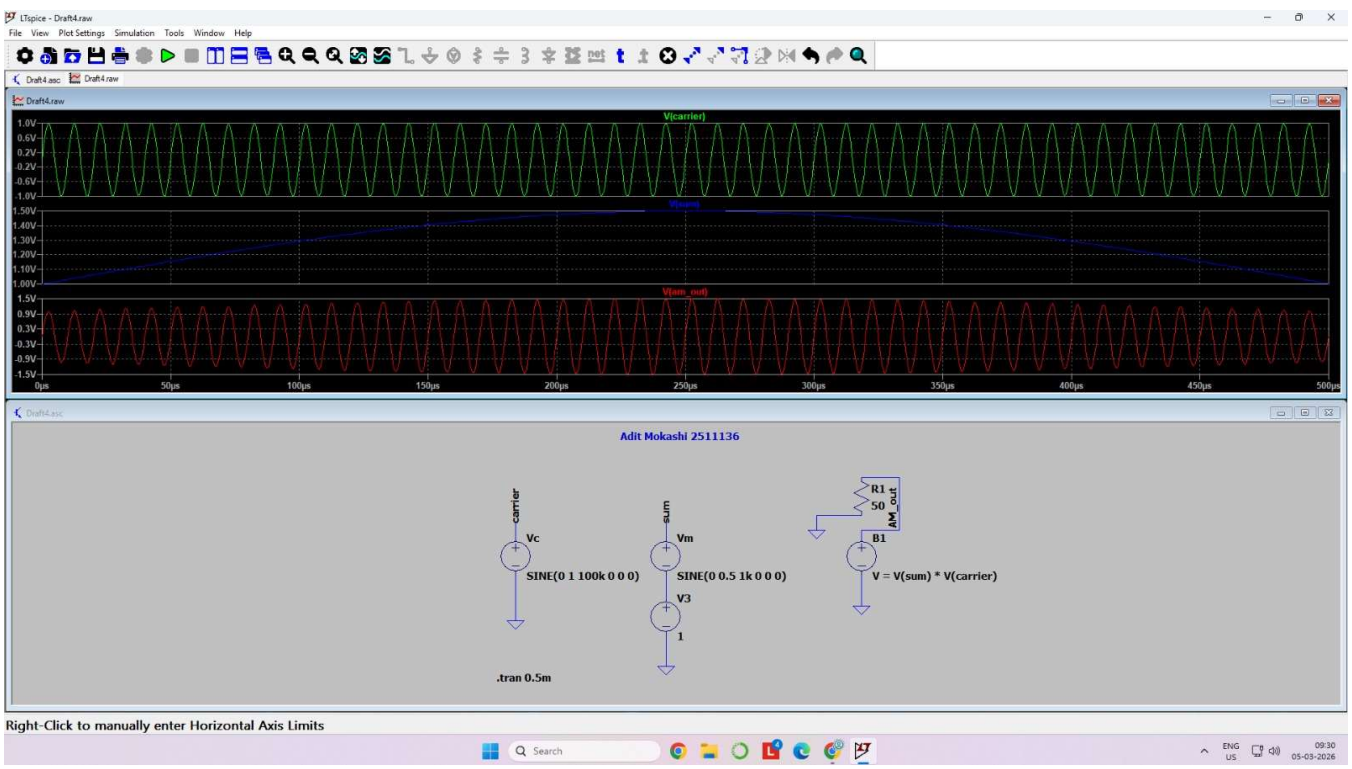
Measuring μ from the Waveform

Measure V_{max} (highest peak) and V_{min} (lowest trough) of the AM waveform:

$$\text{Modulation index } (\mu) = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$$

Observations

Perform the simulation for THREE cases by changing ONLY the Amplitude of Vm each time. Vc (Vamp=1), V3 (1V DC), B1 expression, R1, and simulation time remain unchanged for all cases.



Observation Table

Case	Vm (Am)	Vc (Ac)	Theor. μ	V_max	V_min	Meas. μ	Modulation Type
1	0.5V	1V	0.5	1.5V	0.5V	0.5	Under-modulation
2	1.0V	1V	1.0	2V	0V	1.0	Critical modulation
3	1.5V	1V	1.5	2.5V	-0.5V	1.5	Over-modulation

Conclusion

The AM simulation was performed using four components — Vc (normalised carrier, Vamp=1), Vm (message signal), V3 (DC bias = Ac = 1), and B1 (BV multiplier) — directly implementing the standard AM equation:

$$s(t) = [A_c + m(t)] \sin(2\pi f_c t)$$

- Case 1 (μ=0.5): Clean under-modulated AM. Envelope varied between 0.5 V and 1.5 V.
  Measured μ = 0.5
- Case 2 (μ=1.0): Critical modulation. Envelope minimum just touched 0V.
- Measured μ = 1.0
- Case 3 (μ=1.5): Over-modulation. Phase reversal observed when V(sum) went negative. Measured μ = 1.5